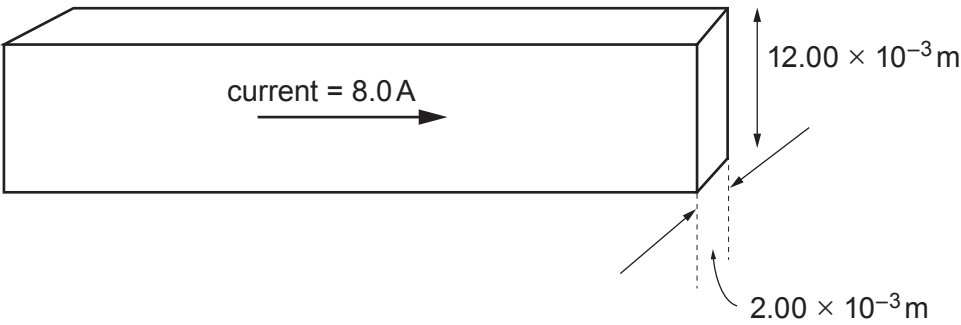


Answer all questions.

1. A strip of metal of rectangular cross-section is shown.



(a) Calculate the drift velocity of free electrons in the strip when there is a current of 8.0 A as shown. The free electron concentration, n , in the metal is $8.5 \times 10^{28} \text{ m}^{-3}$. [3]

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(b) The resistance of the strip is 0.15Ω from end to end. Calculate the energy transferred from electrical potential to thermal in the strip in a time of 20 s, for a current of 8.0 A, and briefly state how the transfer takes place. [3]

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